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303 circuit modelling

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Re: 303 circuit modelling

by [guest](#) » Mon Dec 16, 2024 1:36 am

just wanted to share this project i found, as its both really well done, and exactly what i started out doing with this project. things like this make me happy for a few reasons. the first is thats its just great to see some really technical work that is well executed. the synth sounds good, written in assembly and probably uses every cycle, it was sold for a super cheap price, and the manual and other materials are well designed. second, since someone else already made it, that means i dont have to do it! its not a full 303 model, so im going to continue on with what im doing, but im glad i found it. its such an old project, im suprised i hadnt seen it before, makes me wonder what else im missing.

<https://sonicstate.com/news/2012/09/29/...th-for-29/>

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[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

Re: 303 circuit modelling

by [guest](#) » Wed Dec 18, 2024 1:34 am

today was the envelopes. the VCF envelope is straightforward - acts like an ideal exponential decay. the VCA on the otherhand, quite confusing. its linearish in the beginning, and then does an exponential decay at the end. at first i tried modeling this as a bilinear curve, and then linear + expo, then as an expo with some leakage, but the thing that worked best was a decaying decay rate. what i think is happening, is that the beta of the buffer transistor varies with collector current. so when the envelope first turns on, and there is a lot of collector current, the beta is high, so not much leakage through the transistor. but, as current decreases, beta goes down, and leakage goes up, causing an ever increasing decay rate. i would post a scope shot of 303 vs microcontroller, but the lines are indistinguishable, except for a very brief bit at the beginning where the envelope is a bit flat due to the attack circuit being a bit wierd. at any rate, heres the code:

CODE: SELECT ALL

```
env *= decay;
timer++;
if (timer == 100) {
    decay -= 0.000001;
    timer = 0;
}
output = (uint16_t)(env * 65536);
if (output == 0) {
    env = 1.0;
    decay = 0.999991;
}
```

im casting to an int here for the DAC output, and the timer is needed as floating point has limited resolution. the sample period on the calculations is 200us. the attack and release portions are mostly linear, and quite fast (~1ms).

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[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

Re: 303 circuit modelling

by [guest](#) » Sat Dec 21, 2024 8:24 pm

just a quick update:

1. both VCA and VCF envelopes are complete, and work quite well.
2. i implemented trapezoidal integrators and it started acting funny. better in some ways, worse in others, ill have to dig deeper.
3. i increased the sample rate on the filter core to 200ksp/s and all of my issues went away with the naive integrators, so i might just do that.
4. ive got it doing filter sweeps and VCA modulation, so its making sound as it should.

ive started in on the accent circuit, and am working on a model for that, but once thats done, its should just be a matter of tweaking parameters and writing a sequencer (probably just going to borrow the x0xb0x sequencer).

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[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

Re: 303 circuit modelling

by [Altitude909](#) » Tue Dec 24, 2024 11:50 am

funny that youre still digging into this almost 20 years later.. The last real mystery for me is why do modern duals behave differently than the old ones. i've tried just about every dual transistor on the market and all of them come up short, some better some worse but all come up short compared to the melco parts



[Altitude909](#)
Super Deluxe Wiggler

Posts: 4434
Joined: Wed Aug 24, 2011 5:13 pm
Location: Meesheegan

Re: 303 circuit modelling

by [guest](#) » Wed Dec 25, 2024 12:14 am

yeah, the irony is not lost on me. ive been thinking about writing a book: "all i ever wanted to know about electronics i learned from my 303".

i was actually thinking about that question of yours the other day, as i was digging into a certain part of the filter that made no sense to me, and it turned out that it had to do with the 1583 having such a high beta, and it made me wonder if that was the issue. i seem to recall you having an issue with a x0x that had too much filter warble, and it only went away with a 1583. im beginning to think that the filter warble is actually the 7Hz resonance in the graphs above, and this is mostly determined by the highpass in the diffamp, which in turn is mostly determined by the beta of that 1583, which is over 500.



[guest](#)
Super Deluxe Wiggler

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Re: 303 circuit modelling■ by **guest** » Tue Jan 14, 2025 1:16 am

just wanted to give an update, as the project has slowed a bit. there are a few hundred filter operating points that can not be derived exactly, given the model im using. i could go with cascaded second order stages, which would be able to do them exactly, but would also require more computation time, so ive decided to stick with cascaded first order stages for now. but, this means i need to pick operating points that are as close as i can get them, and in the right areas (accuracy near resonance frequency is my top priority). i tried hand tuning them, and it both takes a long time, and its unclear if its optimal. so today i wrote an algorithm to go through all possible combinations to find the optimal (or at least get me close, and then ill hand tune from there). so far this has been working well, although the algorithm takes a long time to process. ive gotten my hadn tuning down from +/-0.4dB to +/-0.2db error. once ive got it all fine tuned, then i have to use the process on those hundred opeating points. ill probably end up doing a dozen, and then linearly interpolate between them.

im also considering moving over to the RP2350. its cheap and powerful.

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Re: 303 circuit modelling■ by **guest** » Wed Jan 22, 2025 11:37 pm

i feel like my progress on this has slowed, but its always 20% of the time to do the first 80% of the work, and vice versa. i started working on the last filter parameters, and i just wasnt satisfied with the accuracy of the math models. the best i could hope for was +/-0.4dB or so, and when i went back to see what the accuracy was like on some of the middle parameters, it wasnt perfect either. so i went back to the drawing board, and am now trying cascaded second order sallen-key filters, rather than single pole passive filters. this is one extra addition, so minimal computation change, and gives near perfect matching (down to 0.0004dB, and i think thats just due to rounding errors). the math looks good, but i now need to recalculate all the filter parameters, and im not sure how to do that at the moment, as the math is different this time (non-real components need to be accounted for). but, ill figure it out.

the other fun idea i had, was that TI sells codecs with onboard DSPs that implement filters (i think 5 biquad stages?) on both input and output. these can be modified in real time, which means the filter could be handled entirely external to the microcontroller. i wont be doing it for this project, as i want to include the tanh distortion, and that needs to be in the feedback loop, but it might make for an interesting, bare bones synth, where the microcontroller does a simple waveform generation, sends it to the codec for complex and arbitrary filtration, and then to a VCA for gain control (or perhaps use the AGCs on the codec for VCAs). the stereo codecs have 4 of these DSP blocks plus internal analog mixing pathways, so it could get very interesting.

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Re: 303 circuit modelling■ by **tmar92** » Thu Jan 23, 2025 4:07 pm

A lot of stm32 have build-in fixed point multiply accumulate (FMAC) hardware meant for digital filters. That might be useful for this as well?

Re: 303 circuit modelling■ by **guest** » Thu Jan 23, 2025 4:38 pm

im using an M4F from atmel right now, and i think it has an FMAC. i tried measuring the execution times, but the compiler keeps moving my code around on me, so its hard to tell whats actually going on under the hood.

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Re: 303 circuit modelling■ by **tmar92** » Thu Jan 23, 2025 4:52 pm

Aah, i meant a dedicated hardware accelerator for filters that offloads the cpu. You have to program it like peripherals. A quick look at the sam4e data sheet didn't show anything similar, but i could be missing it.

Re: 303 circuit modelling■ by **guest** » Thu Jan 23, 2025 5:02 pm

oh, thats interesting, ill have a look.

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Re: 303 circuit modelling■ by **guest** » Thu Jan 23, 2025 5:07 pm

just had a look, and its 16b fixed point math, which i dont think works for me here. some of the constants are quite small, and the loss of resolution would make it unstable. id also have to rewrite the filters as FIR or IIR, rather than the wonky thing im doing, which would be ok since another process is handling them.

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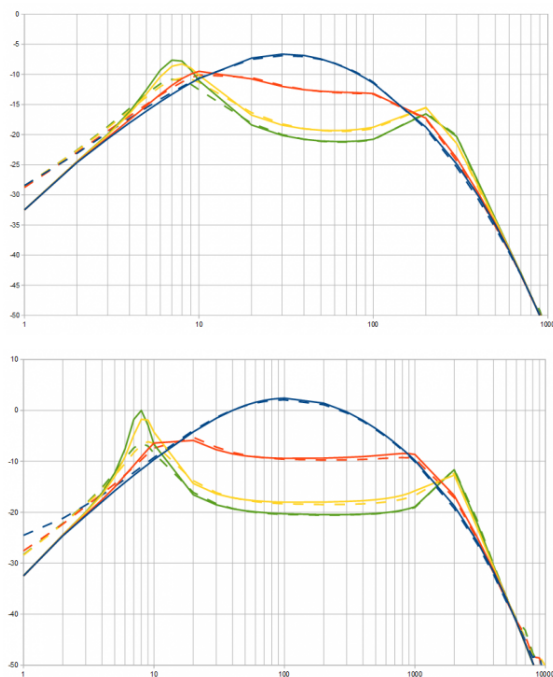
Re: 303 circuit modelling■ by **tmar92** » Thu Jan 23, 2025 5:17 pm

Aah, ok. That makes sense.

Re: 303 circuit modelling■ by **guest** » Fri Jan 24, 2025 3:09 pm

ok, finally feel like im making progress again. the sallen-key filters seem to be working very well. i am getting much closer matching between microcontroller output and measured 303 performance. and i think this also fixes my resonance issues, as the gain mismatch is always the worst near the cutoff frequency, and the gain mismatch is negligible here. it also confirms that the tanh distortion tamps the resonance peaks down to the right level. here are some graphs for lowest and highest cutoff frequencies. solid line is 303, dashed line is microcontroller.

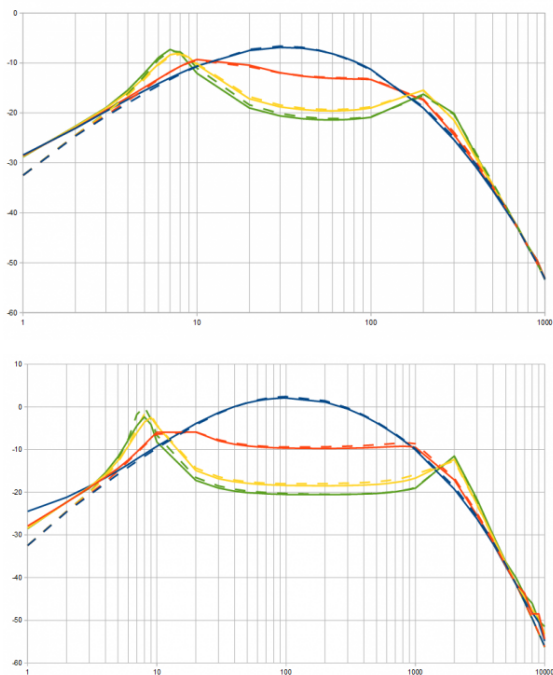
Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am**guest**
Super Deluxe WigglerPosts: 13314
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Super Deluxe WigglerPosts: 13314
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Re: 303 circuit modellingby [guest](#) » Sun Feb 09, 2025 1:14 am

I've been stuck on a few aliasing issues, so to keep making some progress I went back to look at the low frequency bump. At first I didn't think it was too critical, as it's sub-audible, but today I hit the 303 filter with a slow square wave, and that low frequency resonance is really strong. It gave a 400mVpp ring at 7.5Hz and 1.2s ringdown time. You can see the effect when frequency changes, and for constant notes there can be a slight 7.5Hz AM on the signal, as the sharp edges of the waves re-excite the low end. I also realized I had a decent amount of processing time, so adding the third highpass filter in there wouldn't be an issue. Below are plots of the real vs the microcontroller (I have dashed and solid swapped from the graphs directly above - on these dashed is ideal, solid is uC):



There is some slight mismatch, but definitely an improvement over the previous version. I also measured the ringdown of the uC, and set it to 1.2s.

Basically, I removed the old single pole highpass ($ms/(ms+1)$) and added a second order highpass ($as^2/(as^2+bs+1)$) with $a = 0.0156$ and $b = 0.313$. This is approximately an Fc of 8Hz and Q of 0.4. It will probably get tweaked later to deal with the tradeoff of how Q and Fc vary across the various ladder filter cutoff settings.

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Re: 303 circuit modellingby [guest](#) » Fri Feb 14, 2025 7:56 pm

I'm currently working on the accent circuit, which is the last major bit I need to get working. After that, the rest is just details, which isn't nearly as much fun. The accent circuit is actually more complicated than I thought it would be, and I'm going to have to make some approximations for it. But, what I think is interesting, and that I had never known before, is that the accent circuit actually effects the filter, even when accent isn't active. I was trying to track down a weird pulse on the accent->vcf line, and then I realized it was coming from within the vcf itself. When a non-accented envelope signal comes in, it raises the mixing node where the accent envelope also comes in. This backfeeds the accent cap, and raises its voltage, which then stays high for a small bit of time after the envelope has gone away. So it's an additional smoothing on the vcf envelope that builds with a rapid succession of envelope signals. I'm not sure it would be too noticeable for most situations, but interesting none the less.

“



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“



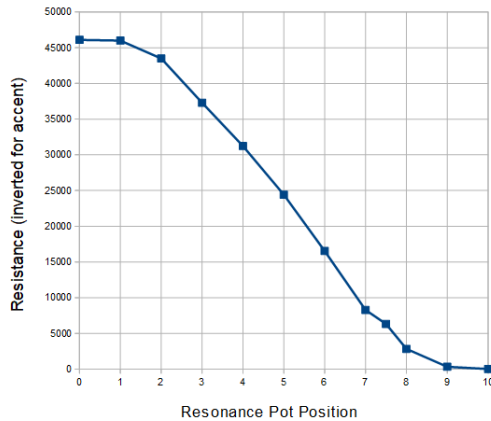
[guest](#)
Super Deluxe Wiggler

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Re: 303 circuit modelling

by **guest** » Sat Feb 15, 2025 2:44 am

after spending a long time trying to figure out why all my filter coefficients were not linearly related to resonance for the accent circuit, i measured the pot taper . . . and, that solves that problem. i dont think im going to model the tapers in the code, as whatever pots are used might have something similar anyways. the accent model is almost done, and i found a way to avoid all of the hairy math that the actual circuit entails.



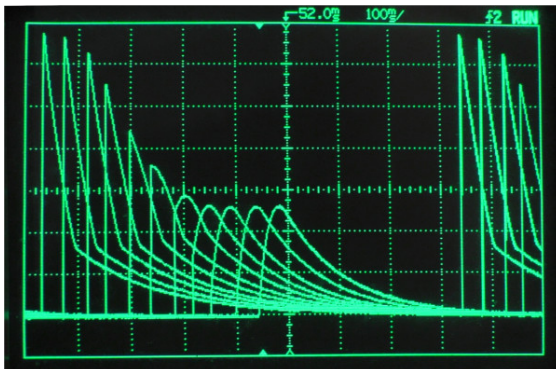
303_reso_pot.png (22.11 KiB) Viewed 1332 times

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Re: 303 circuit modelling

by **guest** » Sat Feb 15, 2025 7:08 pm

and in case anyone is interested, here is the accent signal to the vcf at 11 settings of the reso knob across its range:



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Re: 303 circuit modelling

by **meska** » Sun Feb 16, 2025 2:48 am

👏👏 Impressive work

tremblap wrote:

In theory, there is no difference between theory and practice.
In practice, there is.

https://www.youtube.com/c/Meska_Statik
<https://statikwave.bandcamp.com/>

Re: 303 circuit modelling

by **guest** » Mon Feb 17, 2025 11:53 pm

i decided to stop work on some of the small bits, and try to put what i had together, just to see wehre im at, and if its going to work at all. its been a few months now, and i havent really listened to it yet. so here is a rather long-ish sample where i go through the parameters i currently have working: resonance, cutoff, decay, and envmod:

0:00 / 0:00

303m4_test1.mp3

(3.39 MiB) Downloaded 40 times

there was a depressing half hour or so where it was sounding completely horrible, and then i realized i had my probe to 10x and all of the highs were getting filtered out.

i dont have slide completely running yet (i can tie notes, but the vco does not slide), i do not have the band limited wavetables yet, so this is just a raw saw, which is a bit harsh at times. i do not have the accent yet, although its very close to being done, and i cheated a bit and bumped the resonance to 7% above stock, which you can hear around 3/4 of the way through. in retrospect, perhaps thats a bit much. at any rate, im happy its making sound, but its not where it needs to be. if anyone else has any feedback - ways in which it sounds like it should, and ways in which it does not, im all ears! it will be helpful in figuring out what needs to be tweaked, as im probably a bit close to it at this point.

another fun anecdote: i mentioned a few posts above how the accent circuit was working well, even though i didnt understand why. i now know why - im dyslexic with numbers (which is a thing i knew previously, but...) and i had to subtract 3.26V from the envelope to recenter it for the accent circuit. but, i subtraced 3.62V, which just so happened to be the right about to account for the forward voltage of the diode thats there (which i hadnt been accounting for). in my previous tests, ive found 360mV to be the right amount to average out over a diodes useful range at low currents. i guess i just got lucky this time! every time i tried to add the diode voltage in, the signals got worse, so i just gave up and went with it.

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Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



guest
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“

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guest
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“



meska
Super Deluxe Wiggler

Posts: 1259
Joined: Wed Apr 17, 2013 1:25 pm
Location: France
Contact: [meska](#)

“

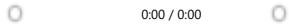


guest
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

Re: 303 circuit modelling■ by **guest** » Tue Feb 18, 2025 12:59 am

that last one just didnt sound right to me, and was making me worried, so went back through the code and found a glaring error in the vcf envmod section. this is much better, oof. also, the noise floor is high as its on a protoboard and im using the 12b onboard DAC with no antialiasing filters. that should all clean up when i get a codec on it.

 303m4_test2.mp3

(3.01 MiB) Downloaded 45 times

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Re: 303 circuit modelling■ by **toneburst** » Tue Feb 18, 2025 12:26 pm

This sounds great @guest! Keep up the great work!!

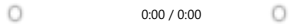
Re: 303 circuit modelling■ by **guest** » Fri Feb 21, 2025 1:32 pm

ok, accents and slides are done, along with the tune knob. todays wtf is that the vca envelope does not trigger until well after the vcf envelope triggers, so you never hear the vcf attack. it can be up to 4ms delayed, and this delay is dependent upon the time since the last note was played. every time i think im about be done, there is another part of the circuit i discover is not as simple as i originally thought it was. after that, its just the vco lookup tables i need to create.

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Re: 303 circuit modelling■ by **guest** » Mon Feb 24, 2025 10:41 pm

ok, so slides and accents are now fully functional, and i fixed some of the noise issues with my setup, so this should be a bit cleaner. still using the 12b onboard DAC.

 303m4_test3.mp3

(3.45 MiB) Downloaded 44 times

i spent a good part of today A/B testing against my 303, and i could get the model to be within 98% or so of the 303 for all the settings i tried. the main issue i am facing now is that the resonance is not consistent with frequency in the model. i added a bit more range to the reso knob on my model so i could tweak to match at the higher frequencies. i also think i have the wightings wrong for how the various modulation sources effect the VCA and VCF. the accent and envmod feel too strong, and the cutoff knob is shifted down a fair bit. but, these are all things i can fix, its the reso issue im stumped on at the moment. the odd bit, is that both the math and microcontroller model match the 303 for bode plots, but ringdown time is all wrong for the microcontroller, and i dont know how that could be. overall, im pretty happy with where its at. if i can fix the reso issue, then i think i can move forward with finishing it off.

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Re: 303 circuit modelling■ by **toneburst** » Tue Feb 25, 2025 5:17 am

Sounds great! Even with the 12-bit DAC. Great work!!

Re: 303 circuit modelling■ by **sphet** » Tue Feb 25, 2025 11:04 am

This sounds great! As a sweng i have done a little bit of audio DSP and have some questions: What environment are you running the emulation in? If the DAC is 12b, what width are your simulating at? float32, float16, fixed16? How are you driving your simulation -- over midi? Thanks for taking us on the journey here it is incredible.

Re: 303 circuit modelling■ by **guest** » Tue Feb 25, 2025 1:52 pm

most of the math is done float16, but i think a lot of it could easily be done in 32b fixed (and some 16b fixed). things like envelopes and such would be fine as int, but the filter core itself most likely needs float16, as the frequency range (and thus coefficients) span a really wide range. im writing it all in arduino (C++) and its running on a feather M4, which is a 150MHz with FPU. right now im using up maybe half of the processing time, although its not evenly distributed. im trying to decide whether to interleave computations or go to a buffered architecture. i know most dsp stuff does the latter, but i like the simplicity of the former. right now the sequences are stored inside of the microcontroller as an array. i just rewrite the code with new notes and slides and such when i want to change it. eventually i will put midi on it, which will make A/B testing a lot easier.

the thing im stuck on at the moment, is that the resonance gets weaker as cutoff is turned up. at first i though this was the standard roll off as you get near nyquist, but im seeing this effect at 2kHz, and sample rate is 50ksp/s with 4x oversampling on the filter. a sweep of the filter doesnt show the reduced resonance, either, but im also plotting in dB, so i wouldnt notice small changes. the ringdown time at 2kHz is half of what it should be, and it gets worse from there. the problem starts around 500Hz. im doing a simple rolling average for my downsampling, and that has some attenuation, but not this much. i need a 10% boost in feedback at 2kHz to get the same response i see at 200Hz. any suggestions are welcomed.

ultimately, i want to make a 303 that is just a microcontroller, and some pots and jacks. or, if it used an external control interface, it could just be a dongle with midi, and audio out.

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Re: 303 circuit modelling■ by **toneburst** » Thu Feb 27, 2025 4:44 am **guest** wrote: 

ultimately, i want to make a 303 that is just a microcontroller, and some pots and jacks. or, if it used an external control interface, it could just be a dongle with midi, and audio out.

Does the Feather M4 have enough I/O and left-over resources to handle a control-surface breaking out all standard 303 synthesis controls?

I wonder if a Eurorack interface board similar to your x0x Heart Pacemaker would be possible, too, thinking ahead.

Re: 303 circuit modelling■ by **guest** » Thu Feb 27, 2025 1:10 pm

“

**guest**
Super Deluxe WigglerPosts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“

toneburst
Super Deluxe WigglerPosts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London

“

**guest**
Super Deluxe WigglerPosts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“

**guest**
Super Deluxe WigglerPosts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“

toneburst
Super Deluxe WigglerPosts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London

“

**sphet**
Common WigglerPosts: 91
Joined: Tue Aug 21, 2018 10:24 pm

“

**guest**
Super Deluxe WigglerPosts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“

toneburst
Super Deluxe WigglerPosts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London

“



i think it has enough, but ill have to double check. a single spi bus could do all the buttons/leds, so its a matter of adc channels. the microcontroller itself should have enough, but its a matter of what they pinned out, or what can be reassigned to the pins they pinned out. right now i have 5 pots on it, which is 1 shy of the usual interface.

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quest
Super Deluxe Wiggler
Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

Re: 303 circuit modelling

by **toneburst** » Fri Feb 28, 2025 6:26 am

Good to know! I can imagine making something really quite small, with just 6 pots, USB-C for MIDI and power, mono audio output on minijack and maybe TRS MIDI in too.

Or I guess you could make the control surface standalone, and have it communicate with the 303 emulation via MIDI. A friend built a Eurorack shield for the OG Axoloti that worked this way, years ago. It had it's own MCU, which handled the pots and buttons, and sent MIDI data to the Axoloti core.

toneburst
Super Deluxe Wiggler
Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London

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